Maximality of the Došen–Petrić Boolean Category

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Abstract

The freely generated zero-identity Boolean category of Došen and Petrić is maximal: every additional equation between distinct parallel arrows, closed under the proof operations and uniform formula substitution, makes the quotient a preorder. This disproves the nonmaximality conjecture in Section 14.2 of the September 2007 revision of *Proof-Theoretical Coherence*. Boolean Coherence reduces a new equation to a contextual equation involving one literal. Uniform substitution extends that equation to every formula. Two structural isomorphisms of Došen and Petrić then force the product projections to coincide. A Lean formalization proves coherence for the equation-generated presentation and verifies maximality, the model corollary, and the occurrence-coordinate constructions. Tagged occurrence sets give coordinates for the reduction and substitution.

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1 Introduction

Došen and Petrić's Boolean proof category admits distinct arrows with the same source and target. Their Boolean Coherence theorem distinguishes these arrows by relations between occurrences of propositional variables. In [1, §14.2, p. 322], they leave open whether the category is maximal and conjecture that it is not. We prove maximality: adjoining any new arrow equation as a substitution-closed schema identifies every pair of parallel arrows.

The proof has two parts. First, a relation distinguishing two arrows contains an edge that can be isolated by composition with a selector and a coselector. This yields the contextual equation used in Došen and Petrić's relative-maximality argument [1, §9.7, pp. 215–216]. Second, we apply two isomorphisms listed in [1, §12.5, p. 294] to turn that contextual equation into equality of the product projections. Their invertibility also follows from the proof of [4, Lemma 7]. This application of the isomorphisms proves the required collapse.

Došen and Petrić prove maximality for cartesian categories [2, §5]. Their proof uses coherence, uniform substitution, and precomposition and postcomposition to reduce a new cartesian equation to equality of the product projections. The corresponding reduction for Boolean arrow equations is proved in Sections 4 and 5.

Earlier results establish maximality for lattice categories and relative maximality for dicartesian categories [3, §§8–10]; these results concern products and coproducts without distribution. Related

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occurrence-selection constructions appear in [4, Lemma 6]. A differing-edge argument also proves maximality for presentations of split preorders, split equivalence relations, and binary relations [5, §16].

Lamarche and Straßburger give a proof-net construction of free categories for their Boolean-category axioms [6, Theorem 3.5.4]. Straßburger studies axioms for $*$-autonomous categories with monoid and comonoid structures [8]. Führmann and Pym interpret cut reduction by an order on arrows and prove soundness and completeness for their net theories [7, Theorems 6.4–6.5]. After precomposition with the inverse right unit isomorphism, their negation triangle [7, (11), p. 28] equates an endomorphism with the identity. The corresponding structural composite defines the zero identity in (2).

We use the occurrence coordinates of [10, §9]: occurrences in a compound formula form tagged disjoint unions, and substitution replaces each occurrence by its own block of occurrences. Section 3 describes these blocks, and Appendix A gives the commutative diagrams relating them to Došen and Petrić's ordinal substitution operation. Section 6 describes the Lean formalization, including the proved inputs to its general maximality theorem. Appendix C records two typographical corrections to the cut-elimination proof in [1]. References to [1] use the printed pagination of its September 2007 revised version.

2 The Boolean proof category

Fix a nonempty set P of propositional variables. Write $\mathrm{Form}(P)$ for the formulas built from P, the constants $\top, \bot$, the binary connectives $\wedge, \vee$, and negation $\neg$. Let $\mathsf{B}(P)$ be the freely generated zero-identity Boolean category defined in [1, §§14.2–14.3]. Its objects are these formulas, and its arrows are the arrow terms of that presentation modulo its equations. We write B when P is fixed. We recall the structure and equations used in the proof.

The connectives $\wedge$ and $\vee$ give binary products and coproducts; $\top$ is terminal and $\bot$ is initial. For $A_1 = A$ and $A_2 = B$, denote the projections and injections by

$$\pi_i^{A,B} : A \wedge B \longrightarrow A_i, \qquad \iota_i^{A,B} : A_i \longrightarrow A \vee B \qquad (i = 1, 2).$$

Pairing and copairing are denoted by $\langle f, g\rangle$ and $[f, g]$, respectively. Write $!_A : A \to \top$ and $\kappa_A : \bot \to A$ for the terminal and initial arrows, and 1_A for the identity of A. Composition $g \circ f$ applies f first. The symbols $f \wedge g$ and $f \vee g$ denote the arrows induced by the product and coproduct bifunctors.

The distributor is an arrow

$$d_{A,B,C} : A \wedge (B \vee C) \longrightarrow (A \wedge B) \vee C.$$

Its first-injection equation is

$$d_{A,B,C} \circ (1_A \wedge \iota_1^{B,C}) = \iota_1^{A \wedge B, C}. \tag{1}$$

This is the coproduct-injection equation of [1, §11.1, p. 234].

Negation is represented by a functor $\neg : \mathsf{B}^{\mathrm{op}} \to \mathsf{B}$, with specified natural double-negation and De Morgan isomorphisms. In particular, write $n_A : \neg\neg A \to A$ for the double-negation isomorphism. The presentation also has arrows $\eta_A : \top \to \neg A \vee A$ and $\epsilon_A : A \wedge \neg A \to \bot$. The *zero identity* is the endomorphism $0_A : A \to A$ defined by

$$0_A = [\kappa_A, 1_A] \circ (\epsilon_A \vee 1_A) \circ d_{A,\neg A,A} \circ (1_A \wedge \eta_A) \circ \langle 1_A, !_A\rangle. \tag{2}$$

The defining zero-identity equation asserts that

$$f \circ 0_A = 0_C \circ g \qquad (f, g : A \to C). \tag{3}$$

Equation (2) is $(0\eta\epsilon)$ of [1, §14.2, p. 317]. Its product and coproduct unit isomorphisms are expressed here in pairing and copairing notation. Equation (3) is the zero-identity law $(0I)$; see [1, §14.3, p. 322].

Remark. Appendix B gives diagrammatic forms of equations (1)–(3), with the source and target of every arrow displayed.

Whenever $\mathsf{B}(A, C)$ is nonempty, define $0_{A,C} = f \circ 0_A$ for any $f : A \to C$. Equation (3) makes this independent of f: both $f \circ 0_A$ and $g \circ 0_A$ equal $0_C \circ g$. Taking $f = g$ gives $f \circ 0_A = 0_C \circ f$. Consequently zero arrows absorb composition whenever the composites are defined: $h \circ 0_{A,C} = 0_{A,D}$ for $h : C \to D$, and $0_{A,C} \circ k = 0_{X,C}$ for $k : X \to A$.

Two arrows are *parallel* when they have the same source and target. A *proof congruence* on B is an equivalence relation on each hom-set preserved by composition and every arrow operation of the presentation, including the derived operations of pairing, copairing, and negation. A *substitution congruence* is a proof congruence also preserved by every uniform substitution $\sigma : P \to \mathrm{Form}(P)$. Here substitution acts on the entire arrow term, including its source and target; this is rule (su) of [1, §2.7, p. 51].

The *total congruence* identifies every parallel pair. A category is *thin*, or a *preorder*, when each hom-set has at most one element. We call B *maximal* when its only substitution congruences are equality and the total congruence. The main result is the following theorem.

Theorem 2.1. *Let P be a nonempty set of propositional variables. The freely generated zero-identity Boolean category $\mathsf{B}(P)$ of Došen and Petrić is maximal. Equivalently, every substitution congruence on $\mathsf{B}(P)$ that identifies two distinct parallel arrows identifies all parallel arrows.*

The proof is completed in Section 5.

3 Occurrences and substitution

For a formula A, write $\mathrm{Occ}(A)$ for the finite set of positions occupied by propositional variables in A. Repeated occurrences of the same variable occupy distinct positions. We use tagged coordinates for these positions.

For finite sets X, Y, let $X \sqcup Y = (\{1\} \times X) \cup (\{2\} \times Y)$ be their tagged disjoint union. The occurrence sets are defined recursively by

$$\begin{aligned} \mathrm{Occ}(p) &= \{*\}, & \mathrm{Occ}(\top) &= \mathrm{Occ}(\bot) = \varnothing, \\ \mathrm{Occ}(\neg A) &= \mathrm{Occ}(A), & \mathrm{Occ}(A \star C) &= \mathrm{Occ}(A) \sqcup \mathrm{Occ}(C) \quad (\star \in \{\wedge, \vee\}). \end{aligned}$$

The label map $\lambda_A : \mathrm{Occ}(A) \to P$ assigns to each occurrence its underlying variable. Distinct positions may have the same label. The left-to-right order gives a bijection from $\mathrm{Occ}(A)$ to $\{1, \ldots, |\mathrm{Occ}(A)|\}$, interpreted as the empty set when the cardinality is zero. This bijection relates the tagged coordinates to the finite ordinals used in [1]; Appendix A.2 expresses the corresponding change of substitution coordinates by a commutative square and a worked example.

Let $\mathbf{Rel}_{\mathrm{fin}}$ be the category of finite sets and binary relations. A relation $R : X \to Y$ is a subset of $X \times Y$; identities are diagonal relations, and composition is relational composition. Transporting the occurrence interpretation of [1] to the tagged coordinates gives a functor $G : \mathsf{B} \to \mathbf{Rel}_{\mathrm{fin}}$ with $GA = \mathrm{Occ}(A)$. For example,

$$G\pi_1^{A,C} = \{((1, x), x) : x \in \mathrm{Occ}(A)\}, \qquad G\iota_1^{A,C} = \{(x, (1, x)) : x \in \mathrm{Occ}(A)\}.$$

The second projection and injection use tag 2. Pairing takes the union of the two relations after tagging their targets; copairing takes the union after tagging their sources. The relation $Gd_{A,B,C}$ is the graph of the rebracketing bijection

$$\mathrm{Occ}(A) \sqcup (\mathrm{Occ}(B) \sqcup \mathrm{Occ}(C)) \longrightarrow (\mathrm{Occ}(A) \sqcup \mathrm{Occ}(B)) \sqcup \mathrm{Occ}(C),$$

which preserves each occurrence of A, B, C [1, §2.9, p. 60]. Zero arrows have empty relations.

Theorem 3.1 (Boolean Coherence, Došen–Petrić)**.** *The functor $G : \mathsf{B} \to \mathbf{Rel}_{\mathrm{fin}}$ is faithful: parallel arrows f, g of B are equal if and only if $Gf = Gg$.*

This is [1, §14.2, p. 321]. In particular, for $p \in P$, the two projections $p \wedge p \to p$ have different occurrence relations, so B is nonthin.

Joyal's collapse theorem concerns cartesian closed categories with a dualizing object; see [1, §14.3, pp. 324–326]. The category B is not cartesian closed. Indeed, faithfulness gives at most one arrow $\top \to E$ for every E, since $\mathrm{Occ}(\top)$ is empty, whereas $1_p \neq 0_p$. An exponential $p \Rightarrow p$ would give a bijection $\mathsf{B}(p, p) \cong \mathsf{B}(\top, p \Rightarrow p)$, a contradiction.

A *signed literal* is p or $\neg p$ for $p \in P$. A formula is in *negation normal form* when negation occurs only immediately before a propositional variable. The double-negation and De Morgan isomorphisms, including the isomorphisms for the truth constants, give an isomorphism $a_A : A \to A^\circ$ to a formula in negation normal form. Its occurrence relation is the graph of the correspondence of positions [1, §14.1, pp. 313–315].

For arrows between formulas in negation normal form, every edge of the occurrence relation joins occurrences of the same signed literal. Indeed, the equivalent presentation of [1, §14.2, pp. 318–321] has signed literals as generators. Its positive structural generators preserve these labels; zero endomorphisms and the arrows η_p and ϵ_p have empty relations. Composition and both tensor operations preserve label agreement. The equivalence with B preserves the occurrence interpretation.

Substitution replaces each occurrence of a variable by a separate block of occurrences of its substituted formula. Appendix A.1 describes these blocks and their label maps by commutative diagrams, following [10, §9]. Let Q be a set of propositional variables and $\sigma : P \to \mathrm{Form}(Q)$ a substitution. If $A[\sigma]$ denotes the result of uniform substitution, there is a canonical bijection

$$\mathrm{Occ}(A[\sigma]) \cong D_{A,\sigma}, \qquad D_{A,\sigma} = \coprod_{x \in \mathrm{Occ}(A)} \mathrm{Occ}(\sigma(\lambda_A(x))). \tag{4}$$

The set on the right consists of pairs (x, t), where x is an old occurrence and t is an occurrence within the formula inserted there. Induction on A proves the bijection: the variable case has one block, the constant cases have none, binary connectives form tagged unions, and negation preserves the occurrence set. In particular, two occurrences of the same variable give two separately tagged copies of its substituted formula.

The organization of these blocks is expressed by a square of sets and functions. Put $E_\sigma = \coprod_{p \in P} \mathrm{Occ}(\sigma(p))$. Define $\mathrm{pr}_D : D_{A,\sigma} \to \mathrm{Occ}(A)$ and $\mathrm{pr}_E : E_\sigma \to P$ by $\mathrm{pr}_D(x, t) = x$ and $\mathrm{pr}_E(p, t) = p$. The function $\widehat{\lambda}_{A,\sigma} : D_{A,\sigma} \to E_\sigma$ sends (x, t) to $(\lambda_A(x), t)$. Thus

$$\begin{array}{ccc} D_{A,\sigma} & \xrightarrow{\widehat{\lambda}_{A,\sigma}} & E_\sigma \\ {\scriptstyle \mathrm{pr}_D}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_E} \\ \mathrm{Occ}(A) & \xrightarrow[\lambda_A]{} & P. \end{array} \tag{5}$$

Both routes send (x,t) to the variable $\lambda_A(x)$ whose substituted formula contains t. The upper arrow forgets which occurrence of that variable supplied the copy; the source $D_{A,\sigma}$ retains it.

Substitution on arrow terms induces a functor

$$F_\sigma : \mathsf{B}(P) \longrightarrow \mathsf{B}(Q), \qquad A \longmapsto A[\sigma], \quad f \longmapsto f[\sigma].$$

Rule (su) makes this functor well defined on arrow terms modulo the defining equations. It preserves identities, composition, and the specified arrow operations. For each arrow, the blocks in (4) describe the occurrence sets of its substituted source and target.

4 Isolating an occurrence

Put $j_A = \iota_1^{A,\top} : A \to A \vee \top$. The contextual equation needed in the proof is

$$j_A \circ \pi_1^{A,\bot} = 0_{A\wedge\bot, A\vee\top}. \tag{K_A}$$

The right-hand side is also $\iota_2^{A,\top} \circ !_\bot \circ \pi_2^{A,\bot}$: its occurrence relation is empty, so Boolean Coherence identifies it with the zero arrow. Thus (K_A) is the contextual equation of [1, §9.7, pp. 215–216].

Lemma 4.1 (Occurrence isolation)**.** *For distinct parallel arrows $f, g : A \to C$ in B, there are a signed literal ℓ and arrows*

$$i : \ell \wedge \bot \longrightarrow A, \qquad r : C \longrightarrow \ell \vee \top$$

such that, after possibly interchanging f and g,

$$r \circ f \circ i = j_\ell \circ \pi_1^{\ell,\bot}, \qquad r \circ g \circ i = 0_{\ell\wedge\bot, \ell\vee\top}.$$

Proof. First replace f, g by $a_C \circ f \circ a_A^{-1}$ and $a_C \circ g \circ a_A^{-1}$. These remain distinct, so it suffices to construct the arrows for the negation normal forms A°, C°. If their selector and coselector are i°, r°, the required arrows for the original formulas are $a_A^{-1} \circ i^\circ$ and $r^\circ \circ a_C$. We may therefore assume that A, C are in negation normal form.

Faithfulness of G gives an edge belonging to exactly one of Gf, Gg. Interchange f, g if necessary, and choose $(x,y) \in Gf \setminus Gg$. The occurrences x and y have the same signed literal ℓ. Set $D = \ell \wedge \bot$ and $E = \ell \vee \top$. For every formula V, the arrows

$$b_V = \kappa_V \circ \pi_2^{\ell,\bot} : D \to V, \qquad t_V = \iota_2^{\ell,\top} \circ !_V : V \to E$$

have empty occurrence relations.

Construct the selector $i : D \to A$ recursively along the tagged path to x. At the selected literal, use $\pi_1^{\ell,\bot}$. At a conjunction, pair the recursively constructed arrow with b_V for the unselected factor, placing it in the order specified by the tag. At a disjunction, compose the recursively constructed arrow with the injection of the selected summand. The displayed rules for G show inductively that Gi consists of the single edge from the sole occurrence of D to x.

Construct $r : C \to E$ dually along the path to y. At the selected literal, use $\iota_1^{\ell,\top}$. At a conjunction, first project to the selected factor and then apply the recursively constructed arrow. At a disjunction, copair that arrow with t_V on the unselected summand. Thus Gr is the single edge from y to the sole occurrence of E.

Relational composition now makes $G(r \circ f \circ i)$ the one-edge relation from $\mathrm{Occ}(D)$ to $\mathrm{Occ}(E)$, whereas $G(r \circ g \circ i)$ is empty. Boolean Coherence gives the asserted equations. □

Proposition 4.2. *If a substitution congruence θ on B identifies distinct parallel arrows, then (K_A) holds modulo θ for every formula A.*

Proof. Lemma 4.1 and compatibility with composition give (K_A) for a signed literal ℓ. If $\ell = p$, let σ substitute A for p and fix the other variables. This yields (K_A) for A: under (4), the literal occurrence in each endpoint becomes the block $\mathrm{Occ}(A)$, and the arrow operations are preserved by F_σ. Appendix A.3 describes this one-occurrence case of the substitution diagrams.

If $\ell = \neg p$, substitute $\neg A$ for p, obtaining the equation for $\neg\neg A$. Transport it along $n_A : \neg\neg A \to A$. Explicitly, let $k_A = j_A \circ \pi_1^{A,\bot} : A \wedge \bot \to A \vee \top$. The product and coproduct equations give the commutative square

$$\begin{array}{ccc} A \wedge \bot & \xrightarrow{\;k_A\;} & A \vee \top \\ {\scriptstyle n_A^{-1}\wedge 1_\bot}\downarrow & & \uparrow{\scriptstyle n_A \vee 1_\top} \\ \neg\neg A \wedge \bot & \xrightarrow[k_{\neg\neg A}]{} & \neg\neg A \vee \top. \end{array}$$

Precomposition and postcomposition of the zero arrow give a zero arrow by absorption. The equation for $\neg\neg A$ therefore gives (K_A) for A. □

5 Contextual collapse

We use the following two isomorphisms to remove the constants from (K_A). Their types occur among the isomorphisms of [1, §12.5, p. 294]; the proof identifies the particular arrows needed here.

Lemma 5.1. *For every formula A, the arrows*

$$q_A = [j_A \circ \pi_1^{A,\bot}, \iota_2^{A,\top}] : (A \wedge \bot) \vee \top \longrightarrow A \vee \top,$$
$$e_A = 1_A \wedge j_A : A \wedge A \longrightarrow A \wedge (A \vee \top)$$

are isomorphisms in B.

Proof. First take $A = p \in P$. By [1, §12.5, p. 294], the source and target of q_p are isomorphic. Each has one occurrence of p, so the relation of any such isomorphism is the unique bijection between these occurrence sets. The arrow q_p has the same relation. Boolean Coherence therefore identifies q_p with that isomorphism.

The same cited list gives an isomorphism $b : p \wedge p \to p \wedge (p \vee \top)$: take $n = m = 1$ in $p^n \wedge (p^m \vee \top) \cong p^{n+m}$ and choose $p^2 = p \wedge p$. The relation Gb is a bijection between two-element occurrence sets. After precomposing b with the product symmetry if necessary, it agrees with the order-preserving bijection Ge_p. Coherence then identifies e_p with an isomorphism. These two identifications also follow from the occurrence-bijection construction in the proof of [4, Lemma 7].

For arbitrary A, substitute A for p and fix the other variables. The functor F_σ sends q_p, e_p to q_A, e_A and preserves their inverse equations. □

Lemma 5.2 (Contextual collapse)**.** *Let θ be a proof congruence on B such that (K_A) holds modulo θ for every formula A. Then B/θ is a preorder.*

Proof. Write $\equiv$ for equality modulo θ. The quotient retains products and coproducts: their pairing, copairing, and defining equations pass to equivalence classes. Fix A, and put $E = A \vee \top$, $S = (A \wedge \bot) \vee \top$, and $s = \iota_2^{A,\top} : \top \to E$. The arrows $0_{A\wedge\bot,E}$ and $s \circ !_{A\wedge\bot}$ have empty occurrence relations and hence are equal in B. Thus (K_A) and the coproduct equations give

$$q_A \equiv [s \circ !_{A\wedge\bot}, s] = s \circ !_S.$$

The inverse equations for q_A hold in the quotient, so

$$1_E = q_A \circ q_A^{-1} \equiv s \circ !_S \circ q_A^{-1} = s \circ !_E.$$

Together with $!_E \circ s = 1_\top$, this makes E terminal in the quotient.

Consequently the projection $r = \pi_1^{A,E} : A \wedge E \to A$ is invertible, with inverse $\langle 1_A, s \circ !_A \rangle$. Lemma 5.1 makes e_A invertible too, and the product equations give $r \circ e_A = \pi_1^{A,A}$. Hence $\pi_1^{A,A}$ is invertible. Its right inverse is the diagonal $\Delta_A = \langle 1_A, 1_A \rangle : A \to A \wedge A$, which must therefore be its inverse. It follows that

$$\pi_2^{A,A} \equiv \pi_2^{A,A} \circ \Delta_A \circ \pi_1^{A,A} = \pi_1^{A,A}.$$

For parallel arrows $f, g : X \to A$, pairing gives $f = \pi_1^{A,A} \circ \langle f, g \rangle \equiv \pi_2^{A,A} \circ \langle f, g \rangle = g$. Thus the quotient is a preorder. □

Proof of Theorem 2.1. Suppose that a substitution congruence θ on $\mathsf{B}(P)$ identifies two distinct parallel arrows. Proposition 4.2 gives (K_A) modulo θ for every formula A. Lemma 5.2 then makes θ total. □

The same argument gives the formulation of maximality in terms of models used in [1, §9.3, p. 194]. An *interpretation* in a zero-identity Boolean category $\mathcal{C}$ assigns an object of $\mathcal{C}$ to each propositional variable and interprets the specified operations; it induces a structure-preserving functor $\mathsf{B}(P) \to \mathcal{C}$.

Corollary 5.3. *Let P be a nonempty set of propositional variables, and let $f, g : A \to C$ be distinct arrows of $\mathsf{B}(P)$. If a zero-identity Boolean category $\mathcal{C}$ satisfies $f = g$ under every interpretation of the propositional variables, then $\mathcal{C}$ is a preorder.*

Proof. Declare two parallel arrows of $\mathsf{B}(P)$ equivalent when all their interpretations in $\mathcal{C}$ agree. This defines a substitution congruence: composing an interpretation with F_σ is another interpretation. The congruence contains $f = g$, so Theorem 2.1 makes it total. In particular, $\pi_1^{p,p} = \pi_2^{p,p}$ holds under every interpretation, for any $p \in P$. Interpret p as an arbitrary object X of $\mathcal{C}$. Its two projections $X \wedge X \to X$ are equal. For any arrows $a, b : Y \to X$, composing $\langle a, b \rangle : Y \to X \wedge X$ with the equal projections gives $a = b$. □

6 Lean formalization

The accompanying Lean 4 development [9] represents formulas and arrow terms by inductive types. An arrow term is indexed by its source and target formulas. The generated equational relation `Derives` defines a quotient category, in which the structural operations and zero-identity construction are proved to satisfy their specified equations. Formula substitution, normalization and the tagged occurrence interpretation are constructed on this syntax.

The theorem `relative_maximality` formalizes the passage from a new substitution-closed equation to totality of the congruence. Its arguments include the structural operations, a faithful occurrence interpretation, normalization, signed-label preservation, substitution and double negation. An alternative contextual-collapse theorem uses the symmetric map

$$h_A = [1_A, 1_A] \circ m_{A,A} : A \wedge A \longrightarrow A,$$

where the mix arrow $m_{A,C} : A \wedge C \to A \vee C$ is given, for any formula B, by

$$m_{A,C} = (\pi_1^{A,B} \vee 1_C) \circ d_{A,B,C} \circ (1_A \wedge \iota_2^{B,C}).$$

This is equation (dm) of [1, §11.1, p. 234]. Its independence of B and the symmetry of mix give $h_A \circ \langle \pi_2^{A,A}, \pi_1^{A,A} \rangle = h_A$. The formal proof uses this identity to derive projection equality from the contextual equations.

For the equation-generated Došen–Petrić presentation, cut elimination and Boolean Coherence are proved in Lean. The development constructs normalization, signed-label preservation, and uniform whole-arrow substitution, and supplies the full presentation as an instance of every hypothesis of `relative_maximality`. The theorem `Presentation.maximality` gives the resulting maximality statement.

The theorem `Presentation.short_maximality` formalizes the proof in Section 5, with explicit inverses for q_A and e_A. The model corollary is proved using full structure-preserving interpretations and their composition with substitution. The formalization also verifies the occurrence-profile and finite-ordinal constructions of Appendix A, the diagram identities of Appendix B, and the concrete source reductions and rank comparisons recorded in Appendix C.

A Substitution in occurrence coordinates

The generalized composition of definable connectives in [1, §2.5, pp. 45–46] records how repeated arguments are substituted into repeated positions. We give its diagrammatic description from [10, §§9.2–9.3]. The diagrams separate the choice of an outer occurrence, the choice of an occurrence within an inserted formula, and the name assigned to that occurrence.

A.1 Profiles and their fibres

For an integer $n \geq 0$, put $\underline{n} = \{1, \ldots, n\}$, with $\underline{0} = \varnothing$. A *formula pattern* on a finite set of placeholders is a formula in those placeholders. Its *occurrence profile* records the placeholder at each position, reading positions from left to right.

Let $U = \{p_1, \ldots, p_k\}$ consist of distinct placeholders and let $M \in \mathrm{Form}(U)$ have m occurrences. Its profile is

$$o_M : \underline{m} \longrightarrow \underline{k}, \qquad o_M(r) = i \quad \text{when position } r \text{ contains } p_i.$$

For each $i \in \underline{k}$, choose a set $Q_i = \{q_{i,1}, \ldots, q_{i,\ell_i}\}$ of distinct local placeholders and a pattern $N_i \in \mathrm{Form}(Q_i)$ with n_i occurrences. The sets Q_i are pairwise disjoint. Write $o_i : \underline{n_i} \to \underline{\ell_i}$ for the profile of N_i. All these numbers may be zero whenever the displayed maps exist.

The occurrences after substitution and the local placeholder indices form, respectively, the sets

$$D = \{(r,t) : r \in \underline{m},\ t \in \underline{n_{o_M(r)}}\},$$
$$C = \{(i,s) : i \in \underline{k},\ s \in \underline{\ell_i}\}.$$

Thus (r,t) specifies an occurrence in the copy of $N_{o_M(r)}$ inserted at position r of M. The pair (i,s) specifies a local placeholder of N_i. Define the injections

$$a_r : \underline{n_{o_M(r)}} \longrightarrow D, \quad a_r(t) = (r,t), \qquad b_i : \underline{\ell_i} \longrightarrow C, \quad b_i(s) = (i,s).$$

There is a unique function $\gamma : D \to C$ for which, for every $r \in \underline{m}$, the following square commutes:

$$\begin{array}{ccc} \underline{n_{o_M(r)}} & \xrightarrow{a_r} & D \\ {\scriptstyle o_{o_M(r)}}\downarrow & & \downarrow{\scriptstyle \gamma} \\ \underline{\ell_{o_M(r)}} & \xrightarrow[b_{o_M(r)}]{} & C. \end{array} \tag{6}$$

The square says that an occurrence inside one inserted copy is labelled by that pattern's profile and then placed in the corresponding target block. Since D is the disjoint union of its source blocks, these squares determine γ on every element. Explicitly,

$$\gamma(r,t) = \big(o_M(r), o_{o_M(r)}(t)\big).$$

Define the block projections $p_D : D \to \underline{m}$ and $p_C : C \to \underline{k}$ by $p_D(r,t) = r$ and $p_C(i,s) = i$. The same construction fits over the outer profile:

$$\begin{array}{ccc} D & \xrightarrow{\ \gamma\ } & C \\ {\scriptstyle p_D}\downarrow & & \downarrow{\scriptstyle p_C} \\ \underline{m} & \xrightarrow[\ o_M\ ]{} & \underline{k}. \end{array} \tag{7}$$

Both routes send (r,t) to $o_M(r)$. This square records which inserted pattern is used; square (6) specifies the map within each such block. Repeated values of o_M give separate source blocks, one for each occurrence of the corresponding outer placeholder.

To identify this construction with actual pattern substitution, replace each $q_{i,s}$ in N_i by (i,s) and denote the resulting pattern over C by $\widetilde{N}_i$. Let $\Phi_0 : U \to \mathrm{Form}(C)$ be given by $\Phi_0(p_i) = \widetilde{N}_i$. Every occurrence of $M[\Phi_0]$ has unique coordinates $(r,t) \in D$. This gives a bijection $\chi : \mathrm{Occ}(M[\Phi_0]) \to D$, and its relation to the label map is the commutative square

$$\begin{array}{ccc} \mathrm{Occ}(M[\Phi_0]) & \xrightarrow[\ \cong\ ]{\ \chi\ } & D \\ {\scriptstyle \lambda_{M[\Phi_0]}}\downarrow & & \downarrow{\scriptstyle \gamma} \\ C & \xrightarrow[\ 1_C\ ]{} & C. \end{array} \tag{8}$$

Indeed, the occurrence with coordinates (r,t) contains precisely the placeholder $\big(o_M(r), o_{o_M(r)}(t)\big)$.

A.2 The finite-ordinal presentation

The finite ordinals used in [1] number the elements of each block consecutively. Put

$$s_0 = t_0 = 0, \qquad s_r = \sum_{q=1}^{r} n_{o_M(q)} \quad (1 \le r \le m), \qquad t_i = \sum_{j=1}^{i} \ell_j \quad (1 \le i \le k).$$

The block encodings are the bijections

$$\begin{aligned} \theta_D : D &\longrightarrow \underline{s_m}, \quad \theta_D(r,t) = s_{r-1} + t, \\ \theta_C : C &\longrightarrow \underline{t_k}, \quad \theta_C(i,s) = t_{i-1} + s. \end{aligned}$$

Define $\delta : \underline{s_m} \to \underline{t_k}$ as the unique function making the following square commute:

$$\begin{array}{ccc} D & \xrightarrow{\ \gamma\ } & C \\ {\scriptstyle \cong}\downarrow{\scriptstyle \theta_D} & & {\scriptstyle \cong}\downarrow{\scriptstyle \theta_C} \\ \underline{s_m} & \xrightarrow[\ \delta\ ]{} & \underline{t_k}. \end{array} \tag{9}$$

The vertical maps change coordinates. The top route keeps the block and local position visible; the bottom route uses a single position number. In coordinates, commutativity says

$$\delta(s_{r-1}+t) = t_{o_M(r)-1} + o_{o_M(r)}(t) \qquad \big((r,t)\in D\big). \tag{10}$$

Every source position belongs to exactly one nonempty block, so this formula determines δ even when some blocks are empty.

In Došen and Petrić's notation, μ and ν_i correspond to o_M and o_i, and $\mu(\nu_1,\dots,\nu_k)$ corresponds to δ. Their block locator selects the unique r for which $s_{r-1} < x \le s_r$; substituting $t = x - s_{r-1}$ in (10) gives their formula [1, §2.5, p. 46]. The square therefore describes the occurrence-profile component of their generalized composition. A pattern can equivalently be specified by erasing its placeholder names to one common symbol and attaching its occurrence profile. In that notation, the generalized composite is represented by the skeleton of the substituted pattern together with the profile δ.

For example, take

$$M(p,q) = q \wedge (p \vee q), \qquad N_1(a,b) = a \wedge b, \qquad N_2(c,d) = d \wedge (c \vee d).$$

The profiles, listed by successive values, are $o_M = (2,1,2)$, $o_1 = (1,2)$, and $o_2 = (2,1,2)$. The source blocks have sizes $3,2,3$, while the target blocks list (a,b) and (c,d). Thus

source position	1	2	3	4	5	6	7	8
placeholder	d	c	d	a	b	d	c	d
δ	4	3	4	1	2	4	3	4

For instance, source position 2 has coordinates $(1,2)$. The profile map gives $\gamma(1,2) = (2,1)$, and the target encoding then gives $\theta_C(2,1) = 3$. Thus $\delta(2) = 3$.

A.3 Shared variables and the maximality argument

The local placeholders may be assigned names in a common variable set Q. Let $\rho : C \to Q$ be that assignment; different elements of C may have the same image. Replace each placeholder $c \in C$ by the variable $\rho(c)$, obtaining the renaming map $(-)[\rho] : \mathrm{Form}(C) \to \mathrm{Form}(Q)$. Define $\sigma : U \to \mathrm{Form}(Q)$ by $\sigma(p_i) = \widetilde{N}_i[\rho]$. Then the following square commutes:

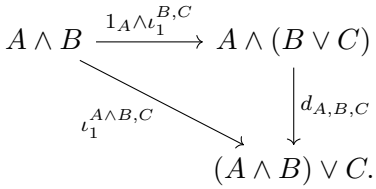
$$\begin{array}{ccc} \mathrm{Form}(U) & \xrightarrow{(-)[\Phi_0]} & \mathrm{Form}(C) \\ {\scriptstyle (-)[\sigma]}\downarrow & & \downarrow{\scriptstyle (-)[\rho]} \\ \mathrm{Form}(Q) & \xrightarrow[1]{} & \mathrm{Form}(Q). \end{array} \tag{11}$$

At p_i, both routes give $\sigma(p_i)$. The constant and connective clauses agree, so structural induction proves commutativity. Renaming changes the labels on the occurrences, while the pairs (r,t) continue to distinguish their positions. This permits the same actual variable to occur in several inserted patterns without identifying their separate occurrences.

For the substitution $p \mapsto A$ in Proposition 4.2, the outer pattern is p, so there is a single source block. This block is $\mathrm{Occ}(A)$. The source and target formulas of the contextual equation are $p \wedge \bot$ and $p \vee \top$; each has one occurrence of p. Their substituted occurrence sets therefore each contain one copy of $\mathrm{Occ}(A)$, and the substituted arrow has type $A \wedge \bot \to A \vee \top$. If the isolated literal is $\neg p$, the substitution $p \mapsto \neg A$ instead gives $\neg\neg A \wedge \bot \to \neg\neg A \vee \top$; the double-negation square in Proposition 4.2 then gives the required equation for A.

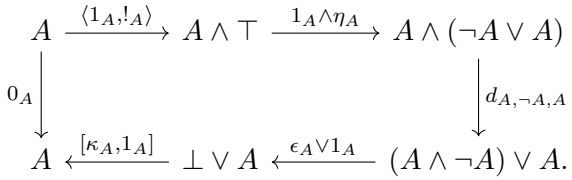
B Diagrammatic forms of the structural equations

The first-injection equation. For formulas A, B, C, equation (1) is the commutative triangle



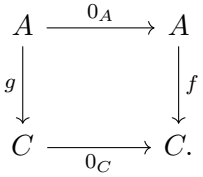
Both paths have source $A \wedge B$ and target $(A \wedge B) \vee C$. The upper arrow is induced by $1_A : A \to A$ and $\iota_1^{B,C} : B \to B \vee C$.

The zero identity. For a formula A, equation (2) is expressed by the commutative diagram



The path across the top, down the right side, and back along the bottom is the five-arrow composite in (2). Its first arrow pairs $1_A : A \to A$ with $!_A : A \to \top$; its last arrow copairs $\kappa_A : \bot \to A$ with $1_A : A \to A$. The arrows between them use $\eta_A : \top \to \neg A \vee A$ and $\epsilon_A : A \wedge \neg A \to \bot$, together with the distributor.

The zero law. For any parallel arrows $f, g : A \to C$, equation (3) is the commutative square



The upper-right path is $f \circ 0_A : A \to C$, and the lower-left path is $0_C \circ g : A \to C$. The arrows f and g need not be equal.

C Two corrections to the cut-elimination proof

The cut-elimination proof in [1, §11.2] contains two typographical errors in its rank comparisons. We use the notation of that proof: r is the rank of the cut being reduced and r' is the rank of a recursive subcut. The left and right ranks are positive; their sum is the cut rank [1, p. 249].

Page 260, case $(w\,\gamma\,3)$**.** In the sentence immediately following the displayed reduction, the inequality $r < r'$ should read $r' < r$. The recursive cuts have the same cut formula and smaller rank: the corresponding one-sided ranks decrease on passing to the indicated premises, while the other one-sided rank is unchanged.

Page 262, case $(w \vee 1)$**.** In the continuation of this case, the phrase "whose rank is 1" should read "whose right rank is 1", as in the analogous argument on p. 261. The selected formula in the right premise is newly introduced by the displayed $\vee L$ rule, so its right rank is 1. Its total rank is the unchanged left rank plus 1, and is therefore at least 2.

Use of AI tools

OpenAI's ChatGPT was used to assist with the development of the proof, preparation of the manuscript, and checks of the arguments and references.

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